VOICERESTOCK AI

Predict. Explain. Call. Reorder.

MASTER PRODUCT & ENGINEERING SPECIFICATION
Για άμεση υλοποίηση με Claude Code

Challenge: Revival — The Intelligent Restocking Agent: Bridging ERPs with AI

Ομάδα: 2–3 full-stack developers | Hackathon: 6–8 Νοεμβρίου 2026

Στόχος: Forecasting stock depletion ανά πελάτη και προϊόν, αυτόματες προτάσεις παραγγελίας στο mock ERP, AI explanation, πρόβλεψη προσαρμοσμένη σε εποχικότητα/προσφορές/αργίες/καιρό και ελληνόφωνος outbound AI Voice Agent που δημιουργεί draft παραγγελίες με φωνητική επιβεβαίωση.

Κατάσταση εγγράφου: Build brief / source-aligned design. Δεν είναι έτοιμος κώδικας, ούτε ισχυρισμός ότι έχουν επαληθευθεί δεδομένα ή APIs που δεν παραδόθηκαν.

Πρωτεύουσα πηγή: 10σέλιδη επίσημη παρουσίαση της Revival που παρέχεται μαζί με το project. Το αναλυτικό data pack (CSV/JSON/mock ERP server/README/submission template) δεν περιλαμβάνεται στο παρόν αρχείο και πρέπει να εντοπιστεί στο workspace.

Σημαντική αρχή υλοποίησης

Ο αλγόριθμος προβλέπει αριθμούς και ημερομηνίες. Το LLM εξηγεί και διαχειρίζεται διάλογο. Ο backend ελέγχει τις ποσότητες, τα δικαιώματα και τα writes στο ERP. Η φωνητική εντολή δημιουργεί draft order — όχι τελική δεσμευτική παραγγελία, διότι αυτή είναι η δυνατότητα που τεκμηριώνεται από το mock ERP.

1. Executive summary & product boundaries

Η DEMO ΔΙΑΝΟΜΕΣ Α.Ε. διαθέτει ιστορικό τιμολογήσεων B2B σε mini markets, καφέ/μπαρ, εστιατόρια και ξενοδοχεία. Το υπάρχον ERP αποθηκεύει τι, πότε και πόσο αγόρασε κάθε πελάτης· η λύση πρέπει να προβλέπει πότε ο πελάτης κινδυνεύει να εξαντλήσει το απόθεμά του και να δημιουργεί στο ERP το κατάλληλο suggested order.

Η λύση αποτελεί middleware μεταξύ mock SoftOne-style ERP και Account Manager. Εξάγει προβλέψεις για όλα τα 1.975 συγκεκριμένα customer-product pairs, δημιουργεί τις προτάσεις, παρουσιάζει τους at-risk πελάτες και πραγματοποιεί προληπτικές συνομιλίες με Account Manager στα ελληνικά.

In scope (MVP + innovation)

- Ingestion, normalization, cleaning και αξιοποίηση δεδομένων SALDOC/MTRLINES/TRDR/MTRL/PRSN/PROMO/holidays.
- Customer × item consumption, depletion date εντός 90 ημερών, confidence/uncertainty, suggested quantity and delivery.
- Seasonality, Greek holidays, promotions, cold-start model με similar customers.
- Weather-aware features με ιστορικές μετρήσεις για training και μελλοντική πρόγνωση για inference.
- LLM explainability grounded σε structured model evidence — κανένας αριθμός από ελεύθερη επινόηση LLM.
- Dashboard “At Risk”, ERP draft writing, Account Manager approval & audit trail.
- Πραγματική εξερχόμενη κλήση σε εγκεκριμένο αριθμό, πλήρης ελληνικός διάλογος, voice modification και voice-confirmed draft creation.
- Forecast evaluation, CSV submission και reproducible demo.

Out of scope για hackathon

- Production ERP financial settlement, committed purchases/payments, τελικό order fulfillment.
- Εγγυημένη πραγματική απογραφή stock στους πελάτες: οι πωλήσεις είναι proxy για κατανάλωση και απαιτούν παραδοχές.
- Ανεπιτήρητες κλήσεις σε απεριόριστους αριθμούς, αμέτρητες αυτόματες παραγγελίες ή ανεξέλεγκτα LLM tool permissions.

2. Source-derived competition requirements (must-have)

Παράμετρος	Επίσημο στοιχείο
Δεδομένα	140 πελάτες, 80 είδη, 24.365 τιμολόγια, 105.757 γραμμές, 34 μήνες (Ιαν 2024–Οκτ 2026), ~€7,96m net sales.
Coverage	1.975 συγκεκριμένα customer × item pairs από prediction_targets.csv.
Data model	SALDOC header και MTRLINES lines· join με FINDOC.
Forecast	Stock depletion date σε ορίζοντα 90 ημερών. Κενό για μη εξάντληση εντός 90 ημερών.
Write-back	Suggested Order JSON βάσει sample_suggested_order.json, POST /api/draft-orders.
UI	At Risk dashboard, ο Account Manager βλέπει/εγκρίνει suggested order.
Submission	Συμπληρωμένο submission_template.csv, suggested orders, repository και live demo.
Αξιολόγηση	Technical 40%, Algorithm accuracy 30%, Business impact 20%, Innovation 10%.
Baseline	Naive MAE 16,1 ημέρες — να βελτιωθεί. Ελλείπον target pair: 60 ημέρες penalty.

Σημείωση: Πηγές: παρουσίαση Revival, διαφάνειες 2–9. Τυχόν διαφορετικά πεδία/συμβάσεις στο data pack υπερισχύουν των ενδεικτικών interfaces αυτού του εγγράφου.

3. Raw data contracts & ingestion rules

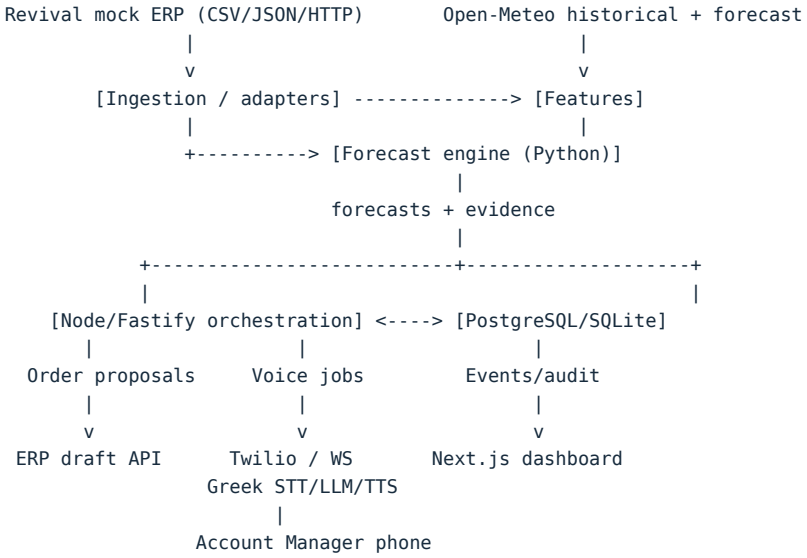
Αρχείο / API	Αναμενόμενη χρήση
SALDOC.csv	Header: FINDOC, TRNDATE, TRDR, SERIES, ISCANCEL.
MTRLINES.csv	Invoice lines: MTRL, QTY1, PRICE, DISC1PRC και FINDOC join key.
TRDR.csv / GET /api/customers	Customer master, segment, location (inspect actual data dictionary).
MTRL.csv / GET /api/items	Product master, product type/unit/category (inspect actual dictionary).
PRSN.csv / GET /api/salesmen	Sales representatives and customer ownership/mapping if present.
PROMO.csv / GET /api/promotions	45 promotions reported; 2 within forecast horizon.
greek_holidays.csv	Greek holidays through 2027.
prediction_targets.csv	Authoritative list of 1.975 pairs.
submission_template.csv	Authoritative column names, order and output format.
mock_erp_server.py	GET /api/saldoc paginated; GET /api/saldoc/{FINDOC}; POST /api/draft-orders.

Cleaning: required exact handling

- Join on FINDOC; deduplicate headers, verify cardinality and orphan lines, reconcile header/lines counts.
- Drop all header ISCANCEL=1 (362 cancelled docs reported), INCLUDING their lines.

- Treat SERIES=7021 as invoice and SERIES=7071 as credit note; credit-note QTY1 is positive in source, so subtract returned quantity from net consumption; document other series discovered.
- Keep zero-price free goods (6.552 lines reported) in physical consumption when they represent real units, but exclude from revenue estimates; test this choice on validation.
- Exclude freight/service lines (2.498 reported) from physical SKU forecasts using actual item master service flag/type — do not invent a field name.
- Use TRNDATE from headers, resolve timezone/date parsing, handle missing values and duplicate records.
- Compute gross positive sales, returns, net quantity and net revenue separately, then aggregate by customer/item/day/week.
- Check aggregation and net-sales reconciliation against sample/reference outputs; record row counts before/after each filtering step.

4. Architecture and service ownership



Service	Ownership	Προτεινόμενη τεχνολογία
web	Dashboard, forecast views, suggestion approval, call transcript/status	Next.js App Router + TS + Tailwind
api	Authorization, business rules, ERP adapter, jobs, voice tools	Fastify + TypeScript + Zod
forecast	Data cleaning, features, validation, inference, submission export	Python + pandas + scikit-learn / CatBoost optional
voice	Outbound Twilio call, persistent WS session, Greek interactive response	Node Fastify WS + Twilio transport
storage	Forecasts, recommendations, call state, audit, idempotency	PostgreSQL; SQLite acceptable for MVP

Repository: apps/web, apps/api, services/forecast, packages/contracts, data/raw (gitignored), data/processed, tests, docs, scripts, submissions. Use pnpm for JS workspace and Python uv/pip for forecast service. Include README with a one-command dev setup where practical.

5. Forecasting: implementation and evaluation

Stepwise model strategy

- Build a reproducible naive baseline that matches the competition definition as closely as documentation permits; store metrics.
- Construct daily/weekly net purchase quantity per customer × item, features for days since last order, cadence, positive purchase quantities, rolling rates (e.g., 30/60/90 days), segment, month, holidays, promotions, history length.
- Forecast demand/burn rate over future days; derive projected depletion date from estimated on-hand stock. Since true client inventory may not be available, explicitly define and validate stock-reconstruction assumptions from transactions/purchase cycles.
- Train a simple, explainable ensemble: segment-aware rolling demand + seasonal profile, optionally gradient-boosted model on robust features. Use time-based validation and compare to baseline.
- For sparse customers, shrink toward peer-group (segment/area/category) priors; report wider uncertainty.
- Generate one forecast per official target pair, even when no depletion within 90 days. Preserve exact template format.
- Rank at-risk pairs and construct delivery date + quantity based on predicted lead time, safety days, pack size/units when known, recent buying pattern, promotions and available constraints.
- Produce auditable per-recommendation evidence object listing inputs and adjustments; never let the LLM manufacture quantities or confidence.

Seasonality / Greek holidays

Features: month, week-of-year, weekday, summer/tourist proxy, Easter/Christmas/holiday proximity and segment-location interactions. Use historical-year alignment; avoid hardcoding that July always increases every product. Apply holiday effects only if measurable, and compare holdout MAE.

Promotions

Join PROMO.csv using actual promotion fields, product/customer scope and valid date ranges. Mark treatment windows, size, pre/post effects and avoid assuming 45 promos each match every customer. Two promotions reportedly lie in forecast horizon; include only applicable ones.

Cold start

The presentation identifies 10 customers starting in 2026. Use segment, location, customer type and product-category nearest peers, hierarchical mean/shrinkage and uncertainty. Do not fit peers using future validation-period information.

Weather-aware features

Map each customer to a valid region/coarse coordinate only if supplied/derivable from permitted fields; use historical weather covering observation period and forecast weather for prediction horizon. Features: Tmax, heat days, precipitation, temperature deviations, interactions per product category/customer segment. Backtests MUST use historical weather as it would have been available at prediction time; historical observed future weather is an oracle and cannot be used as a live-forecast proxy without labeling the resulting metric. Weather feature ablation must demonstrate value; revert if MAE worsens.

Evaluation and leakage checks

- Rolling-origin backtests across several historical cutoffs; split chronologically, not randomly by invoice line.

- Report MAE days, % within 3 days, % within 7 days, no-depletion classification, coverage 1.975/1.975, plus segment/product breakdown.
- Protect future labels, later invoice amounts, future realized promotions/weather and stock reconstruction from leaking into training features.
- Compare baseline → seasonal → promo → cold start → weather; choose best validated method, not most complex.
- If ground truth is available only in scoring environment, report local backtesting explicitly rather than pretending to know official score.

6. Suggested orders and ERP write-back

The slide provides the following illustrative order contract. Actual sample_suggested_order.json and API README are authoritative. Do not hardcode the sample date, which predates the hackathon.

```
{
  "TRDR": 1004,
  "SUGGESTED_DELIVERY_DATE": "2026-11-09",
  "REASON": "Projected stock depletion with seasonal demand",
  "LINES": [{
    "MTRL": 5001,
    "QTY1": 12,
    "PREDICTED_DEPLETION_DATE": "2026-11-12",
    "CONFIDENCE": 0.80
  }]
}
```

The fields TRDR, SUGGESTED_DELIVERY_DATE, REASON, LINES, MTRL, QTY1, PREDICTED_DEPLETION_DATE and CONFIDENCE are slide-supported. The values/dates above are examples only.

Order lifecycle

forecasted → proposed → (voice discussed / dashboard reviewed) → voice-confirmed → server-validated → ERP draft created → visible in dashboard. Failed attempts transition to retryable_failed or needs_review. Final placement beyond draft is OUT OF SCOPE.

- Reconfirm exact customer, product, quantity, unit and delivery before POST; trust a signed account-manager session with allowlisted/verified phone number and an additional PIN or verified session for production-grade authorization.
- Limit quantity, cost/lines and timeframe; block inconsistent units, expired suggestions, customers outside rep scope, duplicate writes and impossible delivery dates.
- Use idempotency key based on authorized suggestion revision + user approval; store ERP response/order id; never announce success until ERP acknowledges it.

7. Fully interactive Greek AI Voice Agent

Proactive outbound call is the UI for an existing recommendation. The voice agent does not invent forecasts. It reads structured evidence and can alter suggested quantities within policy, get an explicit spoken confirmation and then call the ERP draft endpoint.

Outbound triggers and prioritization

- High-severity near stockout (e.g., days-to-depletion less than supplier lead time + safety buffer).
- Weather-driven opportunity or sharp weather change, only if the adjustment is validated and material.
- Applicable promotions, holiday demand shifts and at-risk customer bundles.

- Daily scheduled digest rather than one phone call per SKU; no calls outside working hours, with per-manager cooldown and deduplication.

Transport

Option A: Twilio Programmable Voice → ConversationRelay → persistent WebSocket server → LLM orchestration → Greek TTS / STT configured on supported provider. Validate current Greek language and provider combinations in account before committing. Option B: Twilio bidirectional Media Streams + separate real-time speech-to-speech bridge, if Greek ConversationRelay path fails. No serverless-only WebSocket hosting.

Typed tools — voice cannot access arbitrary SQL or raw ERP credentials

Tool	Allowed effect
get_manager_recommendations	Read assigned at-risk customers and ranked suggestions.
get_recommendation_details	Read structured forecast/evidence, product/qty/depletion/weather details.
explain_forecast	Render concise grounded Greek explanation from evidence.
change_proposed_quantity	Mutate server-side pending suggestion; enforce bounds and pack rules.
confirm_draft_order	Confirm current exact snapshot/nonce in this verified call.
create_erp_draft_order	Server-only action after confirmation and policy validation.
get_draft_order_status	Read ERP-reported result; never claim success early.

Call state machine

```
queued -> dialing -> connected -> greeted -> discussing
 -> quantity_modified? -> explicit_confirmation -> validated
 -> posting_draft -> draft_created | failed | cancelled
 disconnected / timeout -> safe exit, no new order unless posted.
```

Conversation example (synthetic)

AI: «Καλημέρα. Για το Mini Market Ακτή προβλέπεται εξάντληση νερού 500 ml σε τρεις ημέρες. Προτείνω 12 κιβώτια. Να σου εξηγήσω γιατί;»

Manager: «Ναι, γιατί δώδεκα;»

AI: «Η σύσταση προκύπτει από τον ιστορικό ρυθμό, εποχικότητα και διαθέσιμη καιρική πρόγνωση. Η ποσότητα έχει υπολογιστεί από το σύστημα προβλέψεων.»

Manager: «Κάν’ τα δεκαπέντε και φτιάξε παραγγελία.»

AI: «Επιβεβαιώνεις draft για το Mini Market Ακτή, νερό 500 ml, δεκαπέντε κιβώτια, με παράδοση στις [ημερομηνία από engine];»

Manager: «Ναι, επιβεβαιώνω.»

AI (after ERP success): «Το ERP δημιούργησε draft με αναγνωριστικό [επιστρεφόμενο ID].»

Security & voice edge cases

- Phone allowlist, verified identity/session, explicit AI disclosure, opt-in calls, quiet hours, consent to recording if enabled.

- Protect against voicemail, third-party pickup, ASR confusion between “15” and “50”, ambient speech, interruption and disconnection.
- A second clear voice confirmation is required before any write; a single ambiguous phrase cannot directly execute an order.
- Twilio callbacks verified by signature; call jobs idempotent and rate-limited; redact personal information in transcripts.
- Manager only operates customers in own assigned account list. No arbitrary order creation from voice prompt without a current suggestion.
- When uncertain, say what cannot be verified and request re-confirmation or manual dashboard review.

8. Explainability (LLM must be grounded)

Forecast service returns a strict explanation bundle, e.g. product, customer, baseline burn rate, expected depletion date, recommended quantity, applied seasonal multiplier, applicable promotions, peer group (cold start), weather delta, confidence/uncertainty and data quality flags. The LLM only converts that bundle into clear Greek prose and answers questions.

```
type ForecastEvidence = {
  customerId: number;
  productId: number;
  depletionDate: string | null;
  recommendedQty: number;
  confidence: number;
  factors: { name: string; impact: number | null; source: string }[];
  warnings: string[];
  modelVersion: string;
};
```

- No fake explanations: missing effect means “δεν υπάρχει επαρκής ένδειξη”, never fabricated weather/promotion figures.
- Expose contribution attribution only if calculated correctly. Distinguish causal effect from correlation; prefer “συσχετίζεται με” when appropriate.
- Voice response should be short (1–3 sentences), with option for more detail. Do not speak invented availability or confirmed purchase status.

9. API contracts and database schema

Recommended application endpoints (proposed, distinct from official mock ERP API)

Method / path	Action
POST /api/ingest	Read data, validate, build normalized tables.
POST /api/forecasts/run	Run versioned forecasting pipeline, async job.
GET /api/forecasts?customerId=	Forecast results / evidence.
GET /api/risk	Rank at-risk customer/product pairs.
GET /api/suggestions/:id	Suggestion, explanation evidence and state.
POST /api/suggestions/:id/quantity	Modify qty with validation.
POST /api/suggestions/:id/approve	Dashboard approve a draft.
POST /api/suggestions/:id/create-draft	Idempotent ERP draft write.
POST /api/voice/calls	Create authorized outbound call job.
POST /api/voice/twiml	Return Twilio call instructions.
POST /api/voice/status	Twilio signature-verified callbacks.
WSS /api/voice/ws	Persistent voice dialogue transport.
GET /api/reports/submission	Generate exact competition CSV.

Proposed persistence entities

Entity	Important fields
customers / products / sales_reps	Raw ERP keys and permitted metadata; avoid inventing canonical ERP names.
invoice_headers / invoice_lines	Normalized FINDOC, TRDR, TRNDATE, MTRL, QTY, series/cancel fields.
forecast_runs	run_id, cutoff, model_version, config_hash, evaluation_json, created_at.
forecasts	run_id, customer_id, item_id, depletion_date, risk, confidence, evidence_json.
suggestions / suggestion_lines	id, customer_id, delivery_date, version, status, quantity, product_id.
voice_calls	call_sid, manager_id, suggestion_id, lifecycle_status, timestamps.
approvals / audit_events	actor, action, snapshot_hash, outcome, source, correlation_id.
erp_draft_orders	suggestion_id, idempotency_key, ERP response/order id, status.

Require every API write to use typed input validation (Zod), caller authorization, a correlation ID, audit logging and idempotency where relevant. Build a MockERPAdapter that can be replaced with real provider adapters later.

10. Frontend information architecture

Screen	Essential elements
Overview	At-risk customers count, forecast coverage, high-risk SKUs, pending drafts, latest calls.
At Risk	Search/filter by rep, customer segment, region, urgency, depletion date and forecast confidence.
Customer details	Purchase timeline, item forecast, factors for seasonality/promo/weather/cold start, uncertainty.
Suggested Orders	Quantities, delivery, reasoning, state, approval, ERP write status.
Voice Operations	Call queue, status, concise transcript, current suggestion, initiated draft/result.
Evaluation	Baseline vs current MAE, within-3/7 metrics, per-segment breakdown, coverage.

Design priorities: modern professional dashboard, legible tables, weather and promotion chips only where supported, real-time incident/call status, no fake live numbers. For judges expose a “Run demo scenario” action that starts from a fixed reproducible recommendation.

11. Implementation order (24–36 active hours, parallel team)

Phase	Developer 1 / Forecast	Developer 2 / Full-stack	Developer 3 / Voice
0: audit	Inspect data dictionary, template, sample and scoring	Repo setup and mock ERP connectivity	Twilio Greek-language spike
1: data	Clean / join / baseline / tests	Schema + ingestion API + dashboard shell	Working outbound voice + WS
2: model	Seasonal + holidays + promo + cold start	Suggested order generator + ERP draft API	Grounded tools and voice dialogue
3: innovation	Weather experiment + ablation	Realtime dashboard + order lifecycle	Voice quantity modification/confirmation
4: test	1.975 forecasts + MAE + CSV	End-to-end orders + integration tests	Call failures, permissions, voice tests
5: polish	Evaluation plots and limitations	Pitch demo + code README	Phone demo + fallback recording

For 2 developers: Developer A owns data and forecasting; Developer B owns platform and voice. First test a real outbound Greek phone call before investing in voice polish. Forecasting/submission and ERP draft flow remain higher-priority than conversational extras.

12. Tests and acceptance criteria

Data quality

- Zero cancelled invoice quantities remain after filtering; credit note quantities are negative in net demand despite positive source QTY1.

- Freight/service lines excluded; zero-price goods tracked in units if valid stock items.
- No duplicates from header/lines join; data count and financial reconciliation report exists.
- Exactly 1.975 unique forecast pairs; no missing pair; dates and blanks comply with submission template.

Forecast and innovation

- Baseline MAE computed and saved; model MAE compared on leakage-free time-based validation; include within 3/7 days.
- Holiday, promotion, cold-start and weather flags present and tested; ablation documents help/harm.
- Weather forecast for future inference, no future observed weather leakage in past evaluation.
- Confidence reflects actual calibration/uncertainty and is not an LLM-generated arbitrary score.

ERP and voice

- Suggested Order JSON valid against sample and accepted by POST /api/draft-orders; UI shows returned draft ID.
- Calling an allowlisted Greek mobile number succeeds in an actual test; human can ask 2 unscripted Greek questions.
- Manager can say “βάλε 15 κιβώτια”, hear exact confirmed summary, explicitly confirm, and cause exactly one draft write.
- Wrong manager, wrong customer, voicemail, ambiguous quantity, disconnected call or duplicate Twilio webhook cannot create unauthorized duplicate draft.
- If provider is unavailable, dashboard workflow is fully usable; prerecorded fallback clearly labeled in presentation.

13. Environment, setup & deployment

```
# Example environment variable names, NOT real credentials
DATABASE_URL=
ERP_BASE_URL=http://localhost:8080
ERP_API_TOKEN=                  # only if mock server requires it
OPENAI_API_KEY=                 # or selected LLM provider
TWILIO_ACCOUNT_SID=
TWILIO_AUTH_TOKEN=
TWILIO_PHONE_NUMBER=
TWILIO_WEBHOOK_BASE_URL=https://<public-api-host>
TWILIO_VOICE_WS_URL=wss://<public-api-host>/api/voice/ws
WEATHER_API_BASE_URL=https://api.open-meteo.com
VOICE_ALLOWED_NUMBERS=         # server-side allowlist
PUBLIC_APP_URL=http://localhost:3000
DEMO_MODE=true
```

- Never commit secrets, customer identifiers, call recordings or raw sensitive transcripts; use .env.example and .gitignore.
- Host Next.js on Vercel or any web runtime, but the WebSocket voice backend on a persistent Node server (e.g. Render/Fly/Railway) with public HTTPS/WSS.
- Use HTTPS tunnel for local Twilio testing, webhook signature validation, health checks and logging with correlation IDs.
- Create docker-compose for DB + local services if useful. Mock ERP Python server remains its own process on port 8080 unless README says otherwise.
- Keep deterministic demo seed and documented commands: install, start ERP, ingest, train, evaluate, export CSV, run dashboard, trigger call.

14. What Claude Code should do first

CRITICAL: Before implementing, inspect the actual data pack, README, mock_erp_server.py, sample_suggested_order.json and submission_template.csv. The official presentation supplies high-level details but not the full data dictionary. Never guess undocumented field names or API responses. If absent, define adapters/interfaces and mock fixtures clearly marked, then wire them when files are provided.

Build order: (1) inspect filesystem; (2) produce architecture & decisions markdown; (3) implement ingestion and tests; (4) get 1.975-pair baseline with submission output; (5) improve forecasting and track metrics; (6) suggested order JSON/API; (7) dashboard; (8) voice spike; (9) robust interactive voice confirmation; (10) end-to-end demo; (11) documentation.

Paste-ready instruction for Claude Code

```
You are the implementation engineer for VoiceRestock AI.
Read this PDF specification and the official Revival challenge data pack.
Build a runnable TypeScript/Next.js + Python forecasting monorepo.
Treat the official data pack/README/templates as authoritative.
Deliver the mandatory forecast submission for all 1,975 target pairs,
ERP suggested draft orders, At Risk dashboard and reproducible evaluation.
Implement seasonality, Greek holidays, promotions, cold start, LLM-grounded
explanations and carefully validated weather-aware forecasting.
Add a real outbound, conversational Greek AI voice flow for an account
manager, supporting explanation, quantity changes, spoken confirmation,
and exactly-once ERP DRAFT order creation. Never place final orders.
Use typed contracts, tests, authorization, retries, audit and idempotency.
Do not invent source data, official metrics or unsupported ERP fields.
Build in vertical slices, run tests after each slice, and document any gaps.
Start by auditing files and listing assumptions, then implement immediately.
```

15. Demo storyboard & judge-facing message

0–25 sec: Show customer/item at risk and predicted depletion. Explain that the system learned from invoice history (not actual stock sensor data).

25–50 sec: Show evidence: seasonality, holiday proximity, applicable promotion, optional weather adjustment and cold-start treatment if customer is new.

50–95 sec: Trigger phone call to Account Manager. Ask “Γιατί προτείνεις τόσα;”, change quantity by voice, explicitly confirm exact draft.

95–115 sec: Show ERP-created draft ID and updated dashboard; mention idempotency and human control.

115–145 sec: Show honest validation metrics: baseline 16.1 days, your measured MAE, within-3/7, and coverage of 1.975 pairs.

Value claim: “Μετατρέπουμε τα τιμολόγια του ERP σε έγκαιρες, εξηγήσιμες προτάσεις αναπαραγγελίας και επιτρέπουμε στον Account Manager να δημιουργεί draft μέσα από μια πραγματική συνομιλία.”

16. Source traceability & unresolved items

Source slide	Requirement
2	B2B customer stockout, existing ERP invoicing, headers/lines.
3	Four objectives: ingestion, burn rate/depletion, JSON draft, At Risk dashboard.
4	Scale: 24,365 invoices; 140 customers; 80 items; 34 months; seasonality.
5	FINDOC join, series/cancellations, credit notes, free goods, freight services.
6	CSV/JSON data pack and mock ERP endpoints.
7	1,975 target forecasts, submission, suggested orders, demo.
8	Scoring weights, MAE 16.1 baseline, missing-pair penalty.
9	Seasonality, promotions, cold start, LLM explainability.
Added proposal	Weather model and outbound interactive voice order creation.

Open items to inspect, not fabricate: exact submission columns/date format, customer-to-rep mapping, geographic precision, stock estimation method/initial stock, lead time/pack-size rules, promo schema, weather history availability, actual Twilio Greek STT/TTS provider support, mock ERP duplicate behavior and permissions, deploy environment. Clearly surface unresolved issues in README.

Final definition of done: Running repository; accurate/readable README; deterministic scripts; 1,975-row valid submission file; validated models and metric report; At Risk dashboard; ERP draft write-back; interactive Greek phone demo; automated tests; no hardcoded fabricated official scores.